

Nathanael Tesfaye

nathanaeldtesfaye@gmail.com | (240) 527-3776
linkedin.com/in/nathanaeltesfaye/ | github.com/natedoesthings

EDUCATION

Vanderbilt University, School of Engineering - Cumulative GPA: 3.47

Nashville, TN

Bachelor of Science in Computer Science and Mathematics

Expected: May 2026

Relevant Coursework: Data Structures and Algorithms, Programming Languages, Operating Systems, Linear Algebra

TECHNICAL SKILLS

- **Languages:** Java, Python, Swift, C++, C, JavaScript, Kotlin, Racket, Prolog, Ruby, SQL, Bash
- **Frameworks/Libraries:** FastMCP, FastAPI, React.js, Angular, Next.js, CoreML, PyTorch, NumPy

EXPERIENCE

Apple | Swift, React, Python, JavaScript, Docker, Kubernetes, TShark

Cupertino, CA

Software Engineering Intern

June 2025 – Sept 2025

- Built a scalable data ingestion pipeline leveraging MongoDB, AWS S3, and REST APIs, capable of storing and retrieving millions of network packets for large-scale analysis and long-term archival
- Designed and implemented a packet capture parsing service with a companion React-based web client, enabling engineers to visualize and diagnose Wi-Fi network issues five times faster than previous workflows
- Enhanced the team's internal Network Testing application with AI-driven analysis and peer-to-peer connectivity, reducing troubleshooting time for 20,000+ internal users
- Built and optimized native iOS, macOS, and watchOS apps using UIKit, SwiftData, Combine, and Foundation Models

Trane Technologies | C, Ruby, Python, JavaScript, Selenium

Tyler, TX

Software Engineering Intern

May 2024 – Aug. 2024

- Created an application that allows employees to automatically upgrade and downgrade multiple indoor/outdoor units to any version simultaneously cutting time spent on manual labor in half
- Implemented automation scripts using Python to enhance and test system efficiencies in the "Technician App", reducing manual input time by more than 50% and improving system response times by 25%
- Researched and designed the integration of the next-generation user interface framework by analyzing alternatives to help save the company over \$1,000,000/annually in revenue

PROJECTS

GymPeople – Fitness Social Media App | SwiftUI, MapKit, Supabase, CloudFlare R2, Typescript

[Github](#) | [View Project](#)

- Created a SwiftUI client with an MVVM structure to deliver a location-based gym discovery and social profile experience
- Includes features such as posting, commenting, liking, discovering nearby gyms/users, and establishing gym memberships
- Built a robust Cloudflare R2 worker that handles images/documents to help users verify their gym memberships

Tripper – Flight Search Engine | Python, FastAPI, Docker, Railway, SwiftUI, PostgreSQL

[Github](#) | [View Demo](#)

- Implemented full-stack flight coordination engine with a SwiftUI iOS and a FastAPI backend proxying Amadeus APIs via OAuth, enabling secure, paginated flight searches and booking links
- Designed and implemented a scoring algorithm that ranks group flight combinations by arrival/departure time spread
- Parallelized multi-traveler flight searches with debounced airport autocomplete, caching, and a responsive progress indicator

Tic-Tac-Toe AI | SwiftUI, CoreML, PyTorch, NumPy, AVFoundation, Core Data

[AppStore](#) | [View Demo](#)

- Developed a Tic-Tac-Toe application allowing the user to play against a friend or the algorithm
- Utilized PyTorch to generate over 19,000 data points used for the [TicTacToeAI Model](#) via CreateML Tabular Classifier
- Implemented the MiniMax Algorithm to generate accurate data points and the RandomForest Algorithm for training

LEADERSHIP

National Society of Black Engineers | Vice-President

Apr. 2024 – Present

- Led the development and implementation of the chapter's strategic plan, resulting in a 25% increase in membership
- Coordinated with university administration and external partners to secure \$20,000 in sponsorships and grants

Management Leadership for Tomorrow | Software Engineering Fellow

Nov. 2023 – Present

- Accepted into selective 18-month professional development program for high-achieving diverse talent and career guidance